

Problem

Given that $C = -3 + j7$ and $D = 8 + j$, calculate:

(a) $(C - D^*)(C + D^*)$, (b) $D2/C^*$, (c) $2CD/(C + D)$.

Step-by-step solution

Step 1 of 3

Consider the complex numbers, $C = -3 + j7$ and $D = 8 + j$

(a)

The complex conjugate of $D = 8 + j$ is $D^* = 8 - j$.

Determine $(C - D^*)$.

$$\begin{aligned}(C - D^*) &= (-3 + j7 - 8 + j) \\ &= -11 + j8\end{aligned}$$

Determine $(C + D^*)$.

$$\begin{aligned}(C + D^*) &= (-3 + j7 + 8 - j) \\ &= 5 + j6\end{aligned}$$

Calculate the value of $(C - D^*)(C + D^*)$.

$$\begin{aligned}(C - D^*)(C + D^*) &= (-11 + j8)(5 + j6) \\ &= -55 + j40 - j66 - 48 \\ &= -103 - j26\end{aligned}$$

Thus, the value of $(C - D^*)(C + D^*)$ is $\boxed{-103 - j26}$.

Step 2 of 3

(b)

Determine D^2 .

$$\begin{aligned} D^2 &= (8+j)(8+j) \\ &= 64 + j8 + j8 - 1 \\ &= 63 + j16 \end{aligned}$$

The complex conjugate of C is $C^* = -3 - j7$.

Calculate the value of $\frac{D^2}{C^*}$.

$$\begin{aligned} \frac{D^2}{C^*} &= \frac{63 + j16}{-3 - j7} \\ &= \frac{(63 + j16)(-3 + j7)}{(-3 - j7)(-3 + j7)} \\ &= \frac{-301 + j393}{(-3)^2 + (7^2)} \\ &= \frac{-301 + j393}{58} \\ &= -5.19 + j6.776 \end{aligned}$$

Thus, the value of $\frac{D^2}{C^*}$ is $\boxed{-5.19 + j6.776}$.

Step 3 of 3

(c)

Determine CD .

$$\begin{aligned} CD &= (-3 + j7)(8 + j) \\ &= -24 + j56 - j3 - 7 \\ &= -31 + j53 \end{aligned}$$

Determine $C + D$

$$\begin{aligned} C + D &= -3 + j7 + 8 + j \\ &= 5 + j8 \end{aligned}$$

Calculate the value of $\frac{2CD}{C + D}$.

$$\begin{aligned} \frac{2CD}{C + D} &= \frac{2(-31 + j53)}{5 + j8} \\ &= \frac{(-62 + j106)(5 - j8)}{(5 + j8)(5 - j8)} \\ &= \frac{-310 + j530 + j496 + 848}{5^2 + 8^2} \\ &= \frac{538 + j1026}{89} \\ &= 6.045 + j11.53 \end{aligned}$$

Thus, the value of $\frac{2CD}{C + D}$ is $\boxed{6.045 + j11.53}$.